

NATIONAL ECONOMICS UNIVERSITY



DATA-DRIVEN IN MARKETING

FINAL PROJECT :

DATA-DRIVEN CUSTOMER ANALYTICS FOR CUSTOMER RETENTION OPTIMIZATION

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INTRODUCTION

Abstract — This study proposes a data-driven customer management framework to address the challenge of high attrition rates following initial transactions in e-commerce. By leveraging advanced machine learning algorithms combined with in-depth behavioral analysis, the research focuses on accurately identifying high-value customer segments at risk of disengagement. The system's effectiveness is validated through a controlled experimental design, confirming the predictive models' ability to isolate risk factors and ensure forecasting precision. Based on these findings, a multi-tier intervention strategy is established to optimize marketing resource allocation and protect systemic cash flow. The study underscores the importance of shifting the business model from volume-based transaction growth to sustainable customer equity management.

Index Terms — Churn Prediction, Customer Segmentation, Customer Lifetime Value, Machine Learning, Marketing Analytics

BODY

1 Data Description

Dataset provides a comprehensive overview of e-commerce operations throughout the fiscal year 2019 (January 1st to December 31st). It captures 52,924 individual transactions involving 1,468 unique customers. The data architecture is structured across five relational tables, linking customer demographics, transactional behavior, and operational costs.

1.1 Online_Sales.csv

Table 1: Online_Sales.csv

Variable	Data Type	Description
CustomerID	int64	Unique identifier for the customer
Transaction_ID	int64	Unique identifier for each transaction
Transaction_Date	string	Date of the purchase (YYYY-MM-DD)
Product_SKU	string	Unique Stock Keeping Unit ID
Product_Description	string	Detailed name/description of the product
Product_Category	string	High-level category of the product
Quantity	int64	Number of units ordered
Avg_Price	float64	Average price per single unit
Delivery_Charges	float64	Shipping and handling fees
Coupon_Status	string	Indicates if a discount coupon was applied

1.2 Customers_Data.csv

Table 2: Customers_Data.csv

Variable	Data Type	Description
CustomerID	int64	Unique identifier (Primary Key)
Gender	string	Customer gender
Location	string	Geographical location of the customer
Tenure_Months	int64	Total months the customer has been with the platform

1.3 Discount_Coupon.csv

Table 3: Discount_Coupon.csv

Variable	Data Type	Description
Month	string	Month the coupon is valid
Product_Category	string	Category eligible for the discount
Coupon_Code	string	Specific promotional code
Discount_pct	int64	Discount percentage (%)

1.4 Marketing_Spend.csv

Table 4: Marketing_Spend.csv

Variable	Data Type	Description
Date	string	Date of expenditure
Offline_Spend	int64	Spending on TV, Radio, Newspapers, etc.
Online_Spend	float64	Spending on Google, Facebook, etc.

1.5 Tax_Amount.csv

Table 5: Tax_Amount.csv

Variable	Data Type	Description
Product_Category	string	Category name
GST	float64	Goods and Services Tax percentage

2 Exploratory Data Analysis

In the e-commerce environment, consumer behavior is often highly volatile, influenced simultaneously by individual needs, shopping cycles, pricing strategies, and exposure to marketing activities. In a context where customer acquisition costs are steadily increasing, the ability to maintain existing customers is becoming a decisive factor for sustainable revenue growth and the optimization of Customer Lifetime Value (CLV).

With the central objective of optimizing customer retention and providing a quantitative basis for developing personalized retention strategies, we conducted an analysis of the “Marketing Insights for E-Commerce Company” dataset. This dataset comprises 52,924 transactions, integrating both customer-level and transaction-level data. The variables include critical indicators such as Customer ID, transaction time, transaction frequency, order value, online and offline marketing spend, coupon usage behavior, product categories, and metrics reflecting churn behavior.

2.1 Analysis of Revenue Structure and Strategic Product Segmentation

ABC analysis combined with the Pareto principle reveals that the company’s revenue is highly concentrated in a small group of core products. The results indicate that approximately 5% of total SKUs generate nearly 80% of total revenue, reflecting an uneven distribution within the product portfolio.

Specifically, Category A (Core Products) consists of only 3 categories but contributes 78.04% of total merchandise revenue, equivalent to approximately 3.77 million USD out of a total 4.84 million USD in 2019. This demonstrates that current growth relies heavily on a few product lines with high brand recognition and stable market demand.

Conversely, products in Category C represent a large proportion in terms of quantity (consisting of 13 categories – 65% of the total categories) but generate limited economic value (~5.84% of revenue). This phenomenon suggests that the company is maintaining a low-performance portfolio, increasing opportunity costs if marketing resources continue to be spread thinly. In many cases, these may serve as loss leaders to attract traffic; however, current data does not yet show that the company has optimized the conversion process from low-price product groups to those with higher profit margins.

These findings imply that growth strategy should no longer focus solely on transaction volume but must shift toward optimizing Average Order Value (AOV) and maximizing value from medium-performance categories.

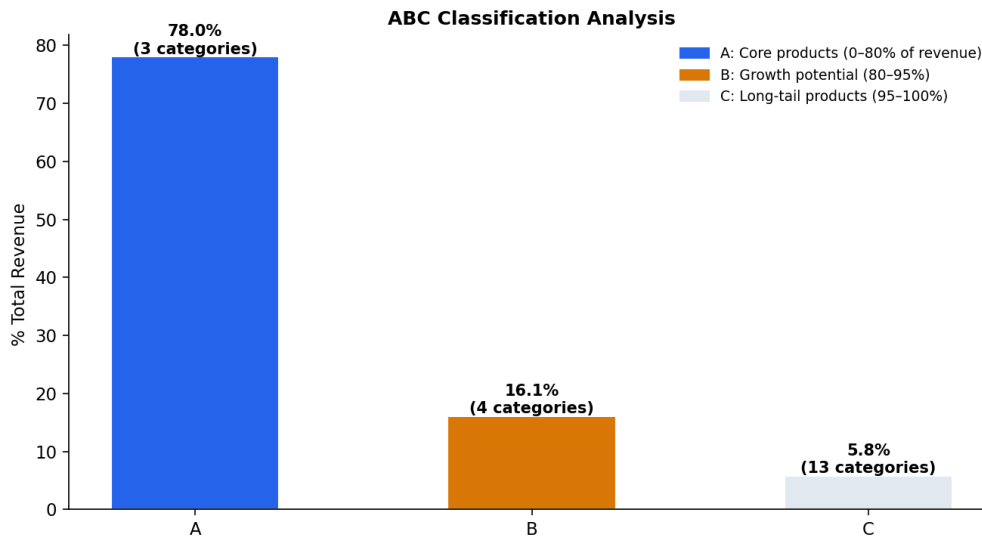


Figure 1: ABC classification distribution

2.2 Revenue Drivers by Sales Channel and Marketing Allocation Efficiency

Order value distribution shows that the majority of transactions are concentrated between 50–200 USD, reflecting the common shopping model in the e-retail industry where customers perform many small-to-medium value transactions.

However, the data shows a significant discrepancy between the mean order value (~215 USD) and the median (~137 USD). This gap reflects the presence of high-value transactions that significantly pull up the average revenue, indicating a right-skewed distribution of revenue.

Correlation analysis between marketing spend and revenue shows that Online Marketing has a correlation coefficient of 0.778, which is higher than Offline Marketing (0.691). This confirms that digital channels serve as a more direct and stable growth driver for sales activities.

Nevertheless, budget allocation analysis reveals a misalignment with actual performance. Marketing expenditure is currently concentrated primarily on offline channels (57–62%), while online channels only account for approximately 38–43%. Furthermore, this ratio has remained relatively stable over time, suggesting that the company is applying a fixed allocation mechanism that fails to react flexibly to performance fluctuations between channels.

A notable point is that marketing spend at certain times does not generate immediate revenue growth. This points to a time-lag effect, meaning there is a certain delay in conversion between the implementation of a campaign and actual customer purchasing behavior.

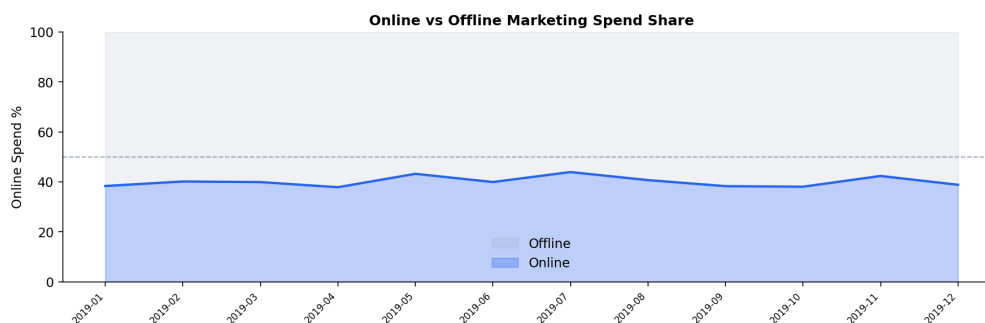


Figure 2: Sales channel distribution

There is a clear basis for the company to reallocate its budget toward increasing the proportion of digital

channels, particularly during periods of high demand or peak shopping seasons.

2.3 Evaluation of Coupon Program Effectiveness

Analysis of coupon usage shows that current promotional programs are having a positive impact on transaction volume. Transactions utilizing coupons account for a large proportion of total sales, indicating that discount programs are effective at stimulating customer participation.

However, when comparing the post-discount invoice value, the data reveals that transactions using coupons have a lower average value than the non-coupon group. This indicates that while coupons help increase transaction volume, they do not boost basket size as expected.

More importantly, after accounting for discount costs, the incremental net revenue per transaction turns negative, suggesting that the current coupon strategy is eroding profit margins rather than creating real value growth.

Category-level analysis shows that the coupon usage rate remains relatively stable at about 33% across most categories. However, Class C product groups, including categories such as More Bags, Lifestyle, and Gift Cards, recorded the highest usage levels, reflecting relatively high price sensitivity.

Notably, core categories in Class A, such as Apparel and Nest-USA, still maintain a coupon usage rate of 33–34%, despite being the primary revenue generators. This suggests the company is at risk of sacrificing profit margins in categories that already possess natural demand.

In contrast, certain categories like Google and Android have usage rates below 28%, reflecting a willingness among customers to pay without the need for price incentives.

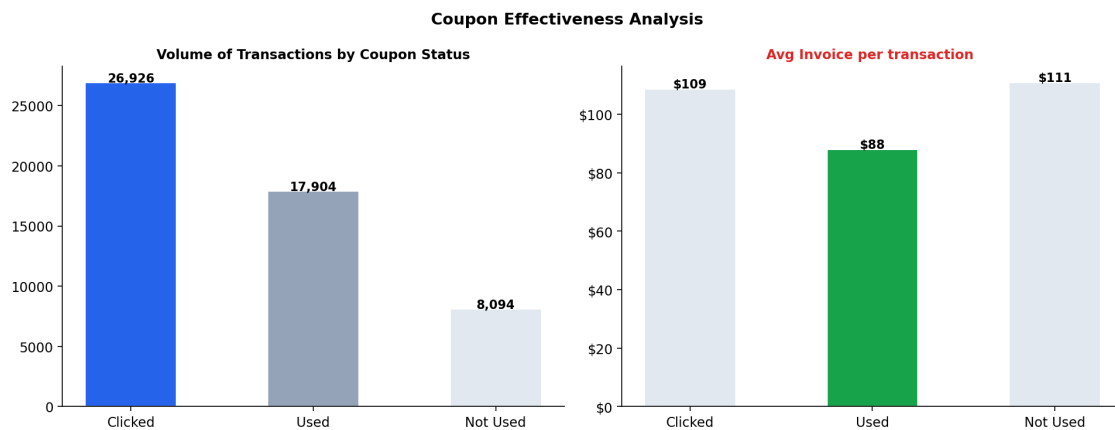


Figure 3: Coupon effectiveness analysis

The company should consider shifting from a blanket discounting model to targeted promotions, prioritizing margin protection in Group A and exploiting cross-selling opportunities in Group C.

2.4 RFM Analysis and Customer Value Tiering

RFM distribution reveals that the current customer base is more engaged than initially assumed.

The Recency variable shows a median of 132 days, meaning 50% of customers have transacted within the last four months. Although a group of customers has been inactive for over 300 days, the majority of the customer base maintains a relatively active level of interaction.

The Frequency variable recorded a median of 11 transactions, indicating that customers do not just make one-off purchases but tend to return frequently. This result refutes the assumption that the company relies primarily on one-time buyers.

Regarding spending value, the Monetary median reached 2,012 USD, and the distribution shows a group of high-value outliers with spending exceeding 10,000 USD. These are customers with high lifetime value who

serve as a strategic revenue source.

RFM tiering shows that Gold customers account for only 17.2% of the total customer base but contribute 50% of revenue, while the Premium group accounts for 21.5% and generates 28.5% of revenue.

Conversely, the Silver (33.7%) and Standard (27.7%) groups make up the bulk of the customer base but contribute only 17.2% and 4.2% of total revenue, respectively.

This reflects a large gap in revenue efficiency per customer and shows that revenue is heavily dependent on a small group of high-value customers.

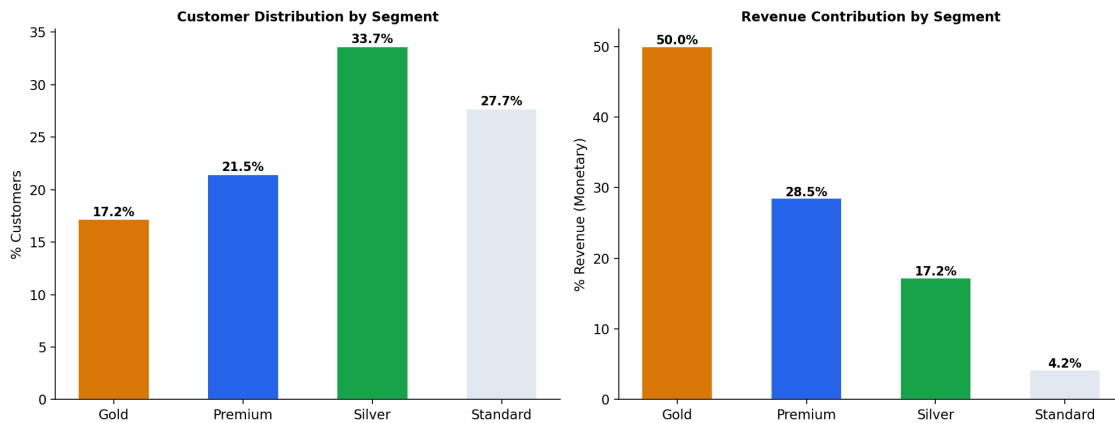


Figure 4: Customer and revenue distribution by segment

2.5 Customer Retention and Churn Risk

The customer retention rate shows a sharp decline immediately after the first purchase. The retention rate drops from 100% to approximately 10% at M+1, representing a nearly 90% decrease in just one month.

After this stage, the retention rate remains relatively stable between 10–15% from M+2 onwards, neither continuing to decline sharply nor achieving significant recovery. Compared to the common e-commerce industry benchmark of 25–30% (according to the Klaviyo E-Commerce Benchmark Report), the entire retention curve currently sits below the 30% threshold, reflecting a serious issue in customer maintenance.

Cohort analysis shows that the 2019-06 group performed best with 15% retention at M+1, while the 2019-01 cohort reached only 6%. Additionally, cohorts from the beginning of the year (2019-01, 2019-02, 2019-03) had large customer volumes but low maintenance quality, suggesting that acquisition efficiency is not yet truly sustainable.

A positive signal is that from M+5 to M+11, retention tends to improve slightly to 15–16%, indicating that customers who survive the first two months have a significantly higher probability of becoming loyal customers. These results suggest that the core problem does not lie in long-term churn, but rather in a failure in second-purchase conversion — the failure to drive customers to perform a second transaction.

2.6 Conclusion and Strategic Direction

The analysis results show that the company's greatest current challenge lies not in generating initial transactions, but in maintaining customers after the first purchase. Cohort analysis shows that the retention rate drops nearly 90% in the very first month (M+1) and remains low in subsequent stages, significantly lower than common e-commerce industry benchmarks. This indicates that the majority of customers currently stop at trial purchase behavior and have not yet converted into loyal customers.

Additional analysis shows that churn does not stem from a single cause but is influenced simultaneously by multiple factors throughout the customer journey. The revenue structure, heavily concentrated in core categories, suggests that the product experience has not been uniformly optimized across all product groups.

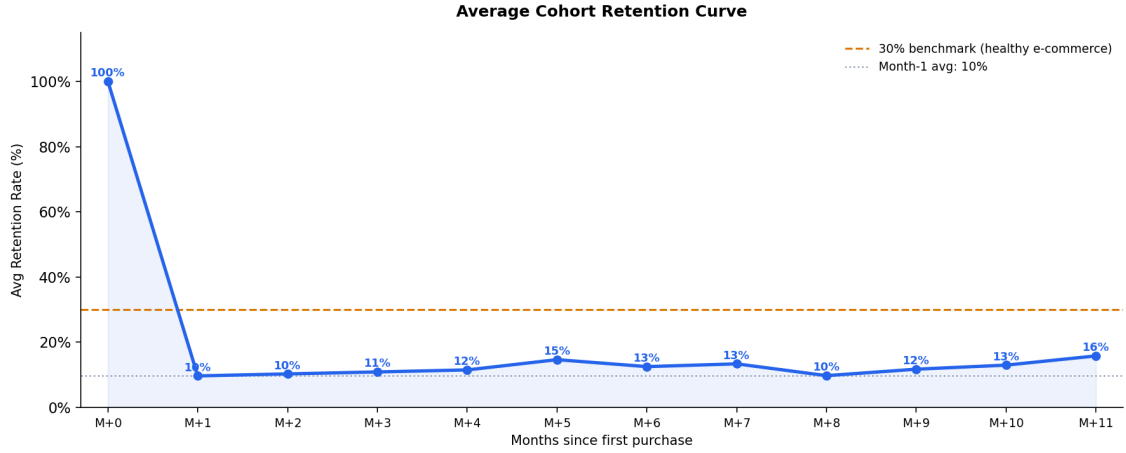


Figure 5: Average Retention by Period

The current coupon program is capable of driving transaction volume but has not yet created added value in repeat purchase behavior, while simultaneously eroding the profit margin per transaction. Furthermore, the imbalance in budget allocation between online and offline channels suggests that the company has not optimized its efficiency in attracting high-long-term-value customer groups.

RFM analysis further indicates that revenue is currently significantly dependent on a small group of high-value customers, while the majority of the customer base remains in segments with low contribution and high churn risk. This reflects a large gap between customer acquisition and customer retention, showing that the company has not maximized the lifetime value of existing customers.

Based on empirical results, the strategic direction for the next phase needs to focus on three main priorities:

First, optimize the first 30 days following the initial transaction. This is the period with the highest churn rate and plays a decisive role in converting first-time buyers into repeat customers. The company should implement automated onboarding flows, personalized email campaigns, and incentive-based reactivation programs to drive the second purchase.

Second, restructure the incentive and customer interaction strategy. Coupon programs need to shift from a blanket discount model to a personalized model based on behavior, aiming to limit margin erosion and increase the likelihood of repeat purchases among potential customer groups.

Third, prioritize the retention of high-value customers. Gold and Premium segments should be managed through loyalty programs, exclusive benefits, and proactive customer engagement to reduce churn risk and maximize Customer Lifetime Value.

Overall, the research results show that the company needs to shift from a growth model based on transaction volume to a retention-driven customer value management model, where the core objective is not only to generate new orders but to build sustainable customer relationships based on actual behavioral data.

3 Modeling

3.1 Feature Engineering

Initial behavioral models have shown that the business is facing a significant challenge in maintaining customers after their first transaction. However, raw transaction data is not sufficient to fully reflect behavioral factors, economic value, or the level of interaction between customers and the business. Feature engineering in this project will transform raw data into analytically meaningful business variables, creating the foundation for steps such as customer segmentation, churn forecasting, CLV analysis, and retention strategy optimization.

3.1.1 Time Features

Shopping behavior in e-commerce is often influenced by temporal factors, including monthly or weekly cycles, or peak shopping periods. To serve the analysis of revenue trends, return behavior, and marketing effectiveness by cycle, time variables are extracted from the **Transaction_Date** field. Specifically, the **Month** variable is created as abbreviated month names (Jan, Feb, Mar...) to serve seasonality analysis and connect with periodic promotional campaigns. The **Date** variable is normalized to a date-only format to support daily data aggregation. Additionally, the **Week** variable is built according to the ISO week standard to serve weekly revenue and marketing trend analysis.

3.1.2 Building Variables Reflecting Coupon and Discount Behavior

In the context of the business implementing many incentive programs to stimulate purchasing behavior, quantifying the level of customer interaction with discount policies is necessary to evaluate the true impact of promotions on retention. Two binary variables have been constructed.

The **Is_Discounted** variable is assigned a value of 1 if the transaction applied a discount (**Discount_pct** > 0) and 0 if there was no promotion. This variable allows the system to determine whether the purchasing decision was influenced by pricing policies.

The **Is_Used_Coupon** variable is created based on the **Coupon_Status**, where a transaction is assigned a value of 1 if the customer actually used a coupon (**Used**) and 0 in all other cases.

3.1.3 Building Actual Invoice Value (Invoice)

One of the limitations of the raw transaction data is that it does not accurately reflect the actual amount paid by customers after accounting for factors such as purchase quantity, selling price, discounts, taxes, and delivery fees. The **Invoice** variable is built to represent the final payment value of each transaction.

Formula:

$$Invoice = (Qty \times Price \times (1 - Disc\%) \times (1 + GST)) + Delivery$$

For transactions using a coupon, the invoice is calculated as the total product value after discount, plus GST tax and delivery costs. For transactions not using a coupon, the system calculates directly on the original price before promotions.

3.1.4 Building Gross Revenue for Category Analysis (Total Revenue)

Beside the final invoice value, the study simultaneously builds a **total_revenue** variable to reflect the actual revenue the business generates from each transaction, serving product category analysis. Unlike **Invoice**, this variable excludes GST tax, as tax is not considered part of the revenue retained by the business but only a collection on behalf of the regulatory authority.

3.1.5 Product Category Classification Based on Pareto Principle (ABC Classification)

To evaluate the business performance of each product category, the study applies the Pareto 80/20 principle to classify categories according to their revenue contribution. Through the **abc_categorizer** function, the system performs the steps: aggregate revenue by category, sort in descending order, calculate the contribution ratio, and calculate cumulative revenue. Based on the cumulative ratio, categories are divided into three strategic groups:

- Group A (0–80%): the core revenue-generating group, which requires inventory priority and restricted deep discounts.

- Group B (80–95%): the group with growth potential, which needs close monitoring for potential upgrade to Group A.
- Group C (95–100%): the low-performance group, suitable for bundling or clearance strategies.

3.1.6 Building the RFM Model (Recency – Frequency – Monetary)

To quantify customer lifetime value and engagement levels, the RFM model is built at the customer level.

The **Recency** variable measures the number of days since the last purchase. A lower value indicates a more active customer.

The **Frequency** variable measures the number of unique transactions the customer has performed. A higher value reflects greater loyalty.

The **Monetary** variable is calculated as the total **Invoice** value, reflecting the actual total spending of the customer.

The reference date is determined as the last day in the data plus one day, ensuring consistency in recency calculation. After that, the system uses `pd.qcut` to divide each variable into 4 scoring groups (1–4), from which the total RFM Score is built with a maximum score of 12. Based on the total score, customers are tiered into four strategic groups:

- Standard (3–5 points): 407 customers
- Silver (6–8 points): 494 customers
- Premium (9–10 points): 315 customers
- Gold (11–12 points): 252 customers

3.1.7 Monthly Marketing Aggregation

To evaluate marketing investment effectiveness over time, the study builds a monthly summary table. The indicators include:

- **Invoice**: total actual monthly revenue
- **Total_Mkt_Spend**: total online and offline marketing costs
- **Pct_Mkt_Spend**: ratio of marketing costs to revenue
- **Pct_Delivery**: ratio of logistics costs to revenue

3.1.8 Features for CLV Prediction and Churn Prediction

Customer Profile Features Create basic features for each customer from transaction data in the calibration period. Group by `CustomerID` and aggregate metrics:

- Base RFM: `first_purchase`, `last_purchase`, `frequency`, `monetary_sum`
- Purchase behavior: `total_quantity`, `unique_categories`, `avg_discount`, `coupon_usage`
- Costs: `delivery_fee`, offline/online spend, marketing spend
- Customer info: `Tenure_Months`, `Gender`, `Location`, `KMeans_Label`

RFM and Recent Features Build behavioral and customer profile metrics to predict CLV. These metrics help segment customers and predict future spend trends, especially to identify who is “hot” or cooling off.

- Recency (**recency_cal**): answer “How long since the last purchase?”, shows recent activity.
- Customer Age (**T_cal**): answer “How long has the customer been active?”, older customers often have higher value.
- Average Order Value (**monetary_avg**): answer “How much does the customer spend per order on average?”, measures order-level value.
- Purchase Frequency (**avg_order_interval**): Average time between purchases, shows whether the customer buys regularly or intermittently.

Recent actions are more important than older history. Customers can change habits over time.

- **freq_last30**: Number of purchases in the last 30 days.
- **spend_last30**: Total spending in the last 30 days.
- **qty_last30**: Total quantity purchased recently.

Momentum Features So far we have full historical purchase data for each customer. But there is one important question: “Is this customer buying more or less than before?”. Momentum features answer that by comparing recent behavior to historical averages.

Purchase Momentum

Formula:

$$\text{Purchase Momentum} = \frac{\text{freq_last30}}{\text{frequency_cal}}$$

Meaning: Ratio of recent purchase frequency to historical average.

- If = 1: customer buys at their usual pace
- If > 1: customer is buying more recently (positive signal)
- If < 1: customer is buying less recently (warning signal)

Spend Momentum

Formula:

$$\text{Spend Momentum} = \frac{\text{spend_last30}}{\text{monetary_sum}}$$

Meaning: Ratio of recent spend to historical spend.

A customer may have strong history (long tenure, high spend), but if momentum is low, they are cooling off. Conversely, a newer customer with high momentum could be a rising star with strong potential.

3.2 Model Customer Segmentation, Customer Lifetime Value and Churn Prediction

3.2.1 Customer Segmentation

Research Objectives and Scope In the context of increasing competition and rising customer retention costs, clearly understanding the shopping behavior of each customer group becomes an important factor in business strategy. Therefore, this research is conducted to segment customers based on actual transaction behavior, thereby supporting the business in building marketing strategies, customer care, and optimizing Customer Lifetime Value.

Model Setup To build the customer segmentation model, the study applies a method combining RFM analysis and the K-Means clustering algorithm.

In the first stage, transaction data is aggregated by each customer via the customer code (`CustomerID`) to build three core metrics representing consumption behavior: Recency (R), Frequency (F), Monetary (M). Proceed to score each metric according to the quartile scoring method. Each metric is divided into 4 score levels from 1 to 4. For Recency, customers with more recent purchase times will receive a higher score because they demonstrate better interaction capabilities. For Frequency and Monetary, customers with higher purchase frequency and larger spending will be assigned higher scores. The total RFM score is determined by the sum of the three components R, F, and M, helping to form a preliminary measure of customer value. Based on this total score, the group initially performs a heuristic classification into groups: Standard, Silver, Premium, and Gold to have a basic business perspective before applying machine learning. K-Means Clustering is deployed with the number of clusters equal to 4. The WCSS (Within Cluster Sum of Squares) value is calculated for many different levels of K from 1 to 10. The Elbow plot results show a clear inflection point at $K = 4$, showing that dividing customers into 4 clusters brings a good balance between detail and interpretability. Based on the average characteristics of the RFM metrics in each cluster, the clusters are interpreted into four main segments including:

- VIP Customers
- Loyal Customers
- Regular Customers
- At-Risk Customers

3.2.2 Customer Lifetime Value Prediction Model

Research Objectives and Scope After completing the customer segmentation step, the research continues to build a Customer Lifetime Value (CLV) prediction model to estimate the total economic value that each customer can bring to the business in the future.

Model Setup To avoid data leakage and ensure practicality, the data is divided chronologically:

- Calibration Period: Contains historical transactions used to create features.
- Holdout Period: The last 90 days, used to determine the actual churn label.
- Recent Window: 30 days before the cutoff date to capture short-term behavioral changes.

The target variable of the model is `actual_clv_90d`, defined as the total actual revenue of each customer in the 90-day holdout. To reduce the influence of outliers, the target value is capped at the 98th percentile, corresponding to 6,642.72 USD. The study tests three regression algorithms including: Random Forest Regressor, Gradient Boosting Regressor, XGBoost Regressor. The parameter optimization process is performed using `RandomizedSearchCV` with 3-fold cross-validation, using MAE (Mean Absolute Error) as the optimization criterion. Experimental results show that Random Forest achieved the best performance and was selected as the final model with an optimal configuration.

Table 6: Performance Comparison of CLV Prediction Models

Model	MAE	RMSE	R^2 Score
Random Forest	502.36	987.61	0.6259
Gradient Boosting	530.93	1014.41	0.6053
XGBoost	533.79	1068.73	0.5619

The Residual Distribution plot shows that most errors are concentrated around 0, proving that the model predicts accurately for ordinary customers and does not show large bias. However, the Actual vs Predicted CLV plot also shows limitations: over-prediction for customers with Actual = 0 and under-prediction for high-value customers (CLV > 4,000 USD). This is a common limitation of Random Forest due to averaging behavior from many decision trees. The ROC-AUC assessment results show that the model is suitable for forecasting overall CLV trends, but should be combined with a churn model to reduce forecasting risk.

3.2.3 Churn Prediction Model

Research Objectives and Scope Customer Churn aims to identify customers at risk of stopping purchases in the future. Early detection of churn signs helps businesses proactively deploy retention strategies, remarketing, and personalized care more effectively.

Model Setup Customers are labeled based on activity during the 90-day Holdout Period.

- Churn (1): No transactions during these 90 days.
- Active (0): At least one transaction occurred.

The model focuses on the following feature groups: engagement features, value features, loyalty & distance features, purchase momentum features and demographic and behavioral features. The study tested: Logistic Regression, Random Forest Classifier, XGBoost Classifier. The Random Forest Classifier achieved the best performance and was selected as the final churn model.

Table 7: Performance Evaluation of the Random Forest Churn Model

Class	Precision	Recall	F1-Score
Active Customers (0)	0.84	0.74	0.79
Churned Customers (1)	0.92	0.96	0.94

Metric	Value
Accuracy	0.91
ROC-AUC	0.9571
Macro Avg F1-Score	0.86
Weighted Avg F1-Score	0.90

Model Evaluation and Conclusion Based on the Random Forest model, the top decision factors include:

- KMeans_Label_1
- Recency_cal
- T_cal
- avg_order_interval

The Random Forest model achieved an ROC-AUC of 0.9571 showing excellent ability to differentiate between churned and retained customers. The model achieved a Recall score of 96%. Based on predicted churn probability, customers are divided into: Risk, Medium Risk, High Risk. Data shows that revenue risk from the High Risk group accounts for approximately 62.81% of total revenue.

4 Strategic Implementation

4.1 Identification of Priority Customers

4.1.1 Post-Forecast Evaluation and Strategic Matrix

The transition from technical predictive models to practical management solutions requires a systematic resource optimization roadmap based on empirical evidence of customer base polarization. Predictive data confirms the existence of a stringent Pareto principle within the research ecosystem, where the high-value segment—comprising only 20% of the population—is projected to contribute 81.1% of total future cash value. This concentration of cash flow creates a massive value gap, evidenced by the fact that the average value of the high-value segment (\$2,156) is more than 255 times higher than that of the standard segment (\$8.45). This reality, combined with the behavioral stagnation of 50% of the lower-tier customers and the thinning of the loyal intermediate segment (\$322), indicates that the organization’s finances are precariously dependent on a small elite group rather than a stable majority.

Furthermore, quantitative analysis of churn probability indicates that 62.8% of total historical revenue currently resides within the high-risk customer group, representing a potential financial loss of up to 2.13 million USD. To address this systemic risk, the study operationalizes predictive results through a two-dimensional segmentation matrix, integrating both predicted lifetime value and churn probability variables to identify strategic touchpoints (see Table 1).

Table 8: Strategic Resource Allocation Matrix Based on CLV and Churn Probability

Customer Segment	Quantitative Characteristics	Core Strategy
High-Value at Risk	High CLV; Churn > 0.7	Urgent Intervention
Loyal VIP	High CLV; Low Churn	Protection & Privileges
Potential	Medium CLV; Low Churn	Stimulate Demand & Upsell
Standard	Low CLV; High Churn	Cost Optimization

4.1.2 Identification of Priority Customers and Financial Impact Assessment

The identification of the target customer group is conducted through a multi-tier filter integrating machine learning metrics, rather than relying on conventional qualitative assessments. Specifically, the system identifies the “High-Value at Risk” segment based on the intersection of three core technical criteria. First is the churn probability extracted from the Random Forest model with a threshold above 0.7. Second is the predicted lifetime value from the XGBoost model, focusing on the top 20% of customers with the highest value. Finally, the momentum index is integrated to identify customers with a clear downward spending trend, reflecting a decline in loyalty despite being primary revenue contributors in the past.

In terms of financial impact, focusing on this segment is vital for budgetary stability. Data analysis identifies that this group represents a revenue at risk of approximately \$498,378, accounting for nearly 88.4% of critical revenue risk. From an investment efficiency perspective, the cost to retain an existing high-end customer is five to seven times lower than the cost of acquiring a new customer of equivalent value. Therefore, the intervention campaign is not merely a marketing activity but serves as a strategic revenue insurance mechanism for the enterprise.

4.2 Experimental Design

4.2.1 Research Hypotheses

To evaluate the real-world impact and validate the system’s predictive power, the study establishes statistical hypotheses to compare the behavior between the identified at-risk group and a control group.

The null hypothesis H_0 is formulated with the assumption that there is no significant difference in retention rates across customer segments. Conversely, the alternative hypothesis H_1 asserts that the urgent retention segment will exhibit a significantly lower retention rate compared to the control group, thereby proving the risk detection capabilities of the deployed machine learning models.

4.2.2 Metrics

The primary metric for the experiment is the retention rate, defined as the percentage of customers who generate at least one transaction within the subsequent 90-day cycle. To ensure comprehensiveness and control for extraneous effects, the study simultaneously monitors guardrail metrics, including average revenue per user and total revenue at risk. Combining these metrics allows for a simultaneous evaluation of both technical predictive power and the practical economic impact of the model.

4.2.3 Statistical Setup

To ensure the reliability and generalizability of the results, the experiment is designed with standardized statistical parameters. The significance level is set at $\alpha = 0.05$ to control for Type I errors, while the statistical power is maintained at $1 - \beta = 0.8$. These settings create a rigorous evaluation framework, ensuring that any recorded behavioral differences are scientifically significant and not due to random fluctuations in the data.

4.2.4 Test Results and Analysis

The execution of the validation test on real-world data reveals an extremely severe polarization between the two groups, confirming that the model has accurately isolated the segment requiring strategic intervention. Detailed results of the testing process are presented in Table 2 below.

Table 9: Validation Test Results for Customer Segmentation

Metric	Control Group	Retention Target Group	Degree of Change
Sample Size	50	378	-
Retention Rate	81.22%	14.90%	-81.7%
p-value	-	-	< 0.0001

The empirical data provides robust evidence for the implementation of management strategies. With a p -value < 0.0001, the study completely rejects the H_0 hypothesis, confirming that the difference in retention rates is absolutely statistically significant. Technically, the fact that the at-risk group has a retention rate 5.4 times lower than the control group demonstrates that the Random Forest and XGBoost models operated effectively in identifying individuals showing signs of churn.

Strategically, a retention rate of only 14.90% in the high-value segment is a “red flag” indicator regarding the loss of strategic assets. This result directly confirms the urgency of implementing a multidimensional intervention framework to rescue the majority of priority customers before disengagement becomes permanent. This serves as the empirical foundation for the enterprise to confidently reallocate resources and budgets toward targeted retention campaigns.

4.3 Intervention Strategy for High-Value at Risk Segment

The segment of customers with high lifetime value but a significant churn probability serves as the top priority in the organization’s retention strategy. Managing this group is executed through a two-stage process, focusing on precise identification and the deployment of multidimensional intervention scenarios.

4.3.1 Identification Framework and Financial Risk Assessment

The identification process for this priority group does not rely on simple qualitative observations but is established through a combination of advanced machine learning models. Specifically, the research integrates results from the Random Forest model (with a churn probability threshold > 0.7) and the XGBoost model to extract the top 20% of customers by predicted CLV. A critical variable included in the filter is the momentum index; when this index turns negative, it reflects a significant decline in transaction frequency and value, even if the customer had a history of deep engagement. Economically, data analysis indicates that this group represents an estimated revenue at risk of approximately \$498,378. Protecting this cash flow is an essential task to prevent budget deficits and ensure financial stability for the business ecosystem.

4.3.2 Multidimensional Intervention Framework and Brand Value Management

To address the complexity of churn behavior in high-value customers, the study proposes a multidimensional intervention framework that transforms technical forecasts into an integrated management strategy. This framework operates on the synergy of four core levels to ensure consistency from analysis to execution.

First, the Analytical Layer serves as the foundational orientation, utilizing the intersection of machine learning models, momentum indices, and ABC classification to precisely determine the target, timing, and intervention tools. Next, the Financial Layer conducts assessments of short-term margin trade-offs to maintain long-term Customer Equity, creating an emergency response mechanism to prevent significant cash flow loss. Simultaneously, the Operational Layer optimizes marketing efficiency by shifting budgets to targeted online channels and personalizing direct communication methods. Finally, the Strategic Value Layer ensures brand sustainability by establishing a transition path from price-based incentives to specific non-financial privileges. The harmonious coordination between these layers not only optimizes ROI but also prevents revenue cannibalization while repositioning brand value in the minds of the elite customer segment.

For high-value customers at risk, the study proposes a flexible intervention strategy that prioritizes a controlled trade-off between short-term profit margins and long-term Customer Equity. Coupled with the execution of the incentive scenario is a shift in marketing budget allocation. Specifically, based on correlation analyses of revenue efficiency, the study recommends shifting 70% to 80% of the total budget to direct advertising platforms such as Google Ads and Facebook Ads. Concentrating financial resources during the year-end peak season (November and December)—when revenue is projected to peak at approximately \$548,000—will leverage natural market demand to increase the effectiveness of retention campaigns.

Within this strategic framework, the use of coupons is strictly controlled based on ABC product classification to protect overall profit margins. Although the general principle is to limit discounts for Category A products (core items accounting for 80% of revenue), for the High-Value at Risk segment, the study proposes a strategic exception: applying personalized discount codes of 20% to 25% specifically for Category A items that the individual frequently purchases. This intervention aims to re-establish engagement through direct financial benefits at the highest-value touchpoints. Simultaneously, communication methods shift from mass channels to direct-to-customer formats as soon as the system records warning signals from the momentum index.

To minimize over-reliance on promotions and protect premium brand positioning, the strategy also proposes a gradual transition from financial incentives to non-financial privilege packages. For customers with a history of price sensitivity, direct discounts will be replaced by value-added benefits such as free express shipping, exclusive gifts, or double reward points. This approach not only helps maintain sustainable loyalty but also prevents revenue cannibalization, ensuring that retention efforts for VIPs at risk contribute positively to the organization's long-term ROI growth goals.

4.4 Business Impact Evaluation and Risk Mitigation Solutions

Based on the empirical results from the urgent retention segment, where the actual retention rate was only 14.9%, the study assesses the financial impact and develops risk prevention measures. Executing this strategy

aims not only to maintain revenue but also to optimize cost structures and protect long-term brand equity.

4.4.1 Expected Business Impact Evaluation

The implementation of data-driven interventions plays a decisive role in the revenue recovery process. With a total revenue at risk of \$498,378, operating the multidimensional intervention framework becomes a financial necessity. The study forecasts that if Category A product incentive policies and specialized VIP care mechanisms help raise the retention rate from 14.9% to a target of 25%, the enterprise is expected to retain approximately \$50,336 to \$62,297 in revenue in the coming quarter. This represents direct profit rescued from a customer group already identified as having a high risk of permanent departure.

Furthermore, this strategy contributes to proactive marketing performance optimization. Validation results confirmed the model's ability to accurately identify segments requiring urgent intervention. Instead of spreading the budget across customer groups that already have natural stability, concentrating 80% of resources on the high-risk group will maximize the conversion rate per unit of cost. Finally, protecting Customer Equity through the presence of the elite group will ensure stability for the entire business ecosystem, providing a solid foundation for upsell and customer upgrade campaigns in the next growth phase.

4.4.2 Operational Risk Mitigation Solutions

To ensure that intervention actions do not cause adverse reactions or harm long-term interests, the study proposes three core risk mitigation solutions:

First, controlling the risk of margin erosion. A common issue is customers generating transactions but passively using discount codes, leading to unnecessary profit loss. To address this, the system only activates personalized discount codes when the purchase momentum index drops to a critical warning level. Additionally, the enterprise should prioritize funnel products or add-on products in the incentive catalog to protect the value of original orders.

Second, managing price sensitivity. To prevent customers from forming a habit of only purchasing during promotions, the study proposes a transition from financial incentives to non-financial privileges. Formats such as free express shipping, exclusive appreciation gifts, or early access to new collections will help retain customers through service experience rather than solely relying on price competition, thereby maintaining premium brand positioning.

Third, mitigating predictive model drift. Customer behavior constantly fluctuates due to seasonality and external impacts, which can lead to forecast deviations over time. The core solution is establishing a continuous feedback loop. All data collected from direct recovery surveys and actual transaction results will be fed back into the processing system to retrain models monthly. This process ensures the accuracy of strategic segments and helps the enterprise adapt promptly to changes in consumer behavior.

CONCLUSION

This research establishes a comprehensive data-driven customer management framework, addressing the core challenge of e-commerce enterprises: severe customer attrition after the first transaction and the risky reliance on a small elite customer segment.

Through analysis and experimentation, the study draws the following key conclusions:

First, regarding the customer ecosystem, the data confirms a stringent adherence to the Pareto principle, where a mere 20% of customers contribute over 81% of the projected revenue value. However, current retention capabilities are at a critical level, with a decline of nearly 90% immediately following the first month. This indicates that the business is over-investing in new customer acquisition while failing to optimize the lifetime value of the existing base, leading to a massive revenue-at-risk figure exceeding 2.13 million USD.

Second, in terms of predictive power and validation, the application of machine learning models such as Random Forest and Extreme Gradient Boosting has delivered superior results. The churn prediction model achieved exceptional accuracy with an ROC-AUC index of 0.9571, allowing the enterprise to precisely identify customers in the process of cooling off interactions. A/B testing results validated this capability by demonstrating a distinct disparity in retention rates between the control group and the at-risk group (81.22% versus 14.90%), providing a solid empirical foundation for subsequent investment decisions.

Third, regarding implementation strategy and economic value, the study proposes a multi-tier intervention framework, shifting from mass marketing to personalized management based on behavioral data. By concentrating 80% of resources on the high-risk priority segment through core category incentive scenarios and experiential privileges, the business is projected to recover between 50,000 and 62,000 USD in revenue per quarter. Furthermore, the establishment of a feedback loop for continuous model retraining ensures the system remains adaptive to market fluctuations.

In conclusion, the research affirms that in the era of big data, a firm's competitive advantage lies not in mere transaction volume but in the ability to understand and protect customer equity. The synchronized implementation—from technical forecasting to strategic intervention—not only secures cash flow but also builds a sustainable growth roadmap, optimizing every touchpoint in the customer journey based on scientific evidence and business reality.

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